

RAGAMALIKA KARUMURI

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EDUCATION

Northeastern University | Master of Science in Information Systems, Boston, MA May 2025
Courses: Database Design, Data Science, Business Analysis, Data Structures and Algorithms, Application Development, Gen AI

Amrita Vishwa Vidyapeetham | Bachelor of Technology in Electrical and Electronics Engineering, India May 2019

TECHNICAL SKILLS

Programming: Python, SQL, R, SAS, Shell Scripting, HTML, CSS, JavaScript
Data Visualization: Tableau, Power BI, Looker, Google Data Studio, Excel (Pivot, Macros, VBA), QuickSight
Database Management: MySQL, PostgreSQL, MongoDB, Snowflake, Oracle, SQL Server
Cloud & Big Data: AWS (S3, EC2, Lambda, Redshift, Glue, SageMaker, Athena), GCP (BigQuery), Databricks, Apache Spark
Tools & Technologies: Jupyter, Git, Postman, Grafana, Figma, Balsamiq, Airflow, dbt, Kafka, Docker, Jenkins, Kubernetes
Certifications: Google Analytics, AWS Cloud Practitioner, AWS Solutions Architect Associate, Generative AI by Microsoft

PROFESSIONAL EXPERIENCE

Humanitarians AI, Boston, MA June 2025 – Present
Generative AI Engineer

- Designed and deployed RAG pipelines with LangChain, OpenAI API, and Pinecone to retrieve insights from structured datasets, reducing manual data lookup time by 65%
- Fine-tuned domain-specific LLMs and implemented prompt engineering, improving accuracy by 30% while reducing inference latency by 40% through via optimized workflows

Northeastern University, Boston, MA Jan 2025 – May 2025
Teaching Assistant – Business Analysis & Information Engineering

- Mentored 60+ graduate students in data storytelling, A/B testing, and Agile collaboration using Slack, Zoom, EasyRetro, and Lean Coffee, boosting participation and enhancing cross-functional teamwork
- Delivered guidance using industry frameworks (Lean Change, Six Sigma) and facilitated AI strategy & digital transformation discussions, simplifying complex concepts for innovation-driven decision-making

OneData Software Solutions, New York, NY Sep 2024 – Dec 2024
AWS Cloud Intern

- Built POC data pipelines using AWS Glue, Lambda, and Kinesis for batch and streaming sources into an AWS Data Lake on S3 and enabling self-service analytics in Athena and Redshift
- Developed a HIPAA-compliant ELT framework with AWS Glue and Terraform to automate ingestion, transformation, and data cataloging, reducing setup time by 40%
- Prototyped predictive analytics models in Amazon SageMaker and visualized outcomes in QuickSight dashboards powered by Redshift datasets, improving model accuracy by 15%
- Created observability dashboards in CloudWatch and Grafana to monitor SLA adherence, pipeline health, and cost optimization metrics in real time

Infosys Limited, Hyderabad, India Jan 2020 - May 2022
Senior Data Engineer

- Implemented **ETL** pipelines using Apache Airflow, DBT, and Python to process multi-terabyte POS transaction data, integrating Kafka, reducing data latency from 24 hours to under 2 hours
- Developed data cleaning, transformation, and normalization frameworks in PySpark and **SQL**, ensuring 99% data quality and regulatory reporting, while automating validation checks through SAS scripts
- Optimized complex analytical queries in Snowflake and PostgreSQL by implementing indexing, partitioning, and query refactoring, reducing execution time by 60%
- Delivered Power BI and Tableau dashboards to monitor retail **KPIs** and enabling real-time trend visibility and reducing reporting turnaround time by 60%
- Automated deployment pipelines with Jenkins CI/CD, containerized with Docker and orchestrated on Kubernetes, enabling high scalability and reliability

ACADEMIC PROJECTS

MBTA AI chatbot | LangChain, OpenAI LLMs, Pinecone Vector DB, Snowflake, Kafka, Docker, Airflow, Whisper, Next.js

- Engineered a full-stack RAG pipeline integrating real-time transit data with automated streaming and speech processing, achieving 95%+ query accuracy through a responsive chatbot interface

Chronic Kidney Disease Prediction | Python, scikit-learn, SVM, KNN, Voting Classifier, Random Forest, EDA

- Implemented an ML model for CKD prediction, achieving 98% accuracy through preprocessing, exploratory analysis, engineered features, and ensemble methods, with results communicated via detailed visualizations and insights